2026-03-14

Stack Overflow Users: Their Background and How They Work

Damian Antos

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OUTLINE

- Executive Summary
- Introduction
- Methodology
- Results
  - Programming languages and databases trends
  - Dashboard
- Discussion
  - Findings & Implications
- Conclusion
- Appendix

EXECUTIVE SUMMARY

- Analysis of the Stack Overflow Developer Survey shows that **JavaScript remains the most widely used programming language**, followed by SQL and Python.
- In the database category, **PostgreSQL and MySQL are the most commonly used systems**, while PostgreSQL is also among the most desired technologies for future work.
- Among cloud platforms, **Amazon Web Services (AWS), Microsoft Azure, and Google Cloud** are the most widely used solutions.
- The demographic analysis shows that **most developers are between 25 and 44 years old** and a large proportion hold a bachelor's or master's degree.
- These results highlight key technology trends in the developer ecosystem and may help developers choose technologies to learn and companies understand current industry demand.

INTRODUCTION

Nowadays, the developer ecosystem includes a wide range of technologies. Some of them are widely used in the industry, others are rapidly gaining popularity, while some are gradually losing relevance. However, how do these trends look when we analyze real data?

This report may be useful for people working in the technology industry as well as for those considering a career change. By analyzing data from the Stack Overflow Developer Survey and job market information, this report provides insights into current technology trends.

The analysis focuses on several key questions: which technologies are the most widely used by developers, which technologies developers would like to work with in the future, which technologies offer the highest earnings, and whether the demand for certain technologies in job postings corresponds to the number of specialists working with them.

METHODOLOGY

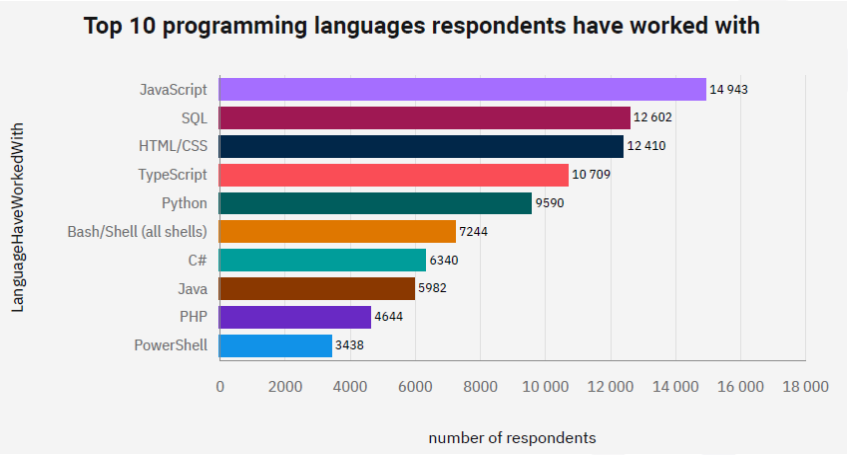
- Data Sources:
  - Stack Overflow Annual Developer Survey 2024 (CSV file)
  - Job postings from Naukri.com (collected via API, JSON format)
  - Programming languages dataset (web scraped)
- Data collection methods:
  - Downloading and loading CSV data
  - Collecting data via API (Naukri.com)
  - Web scraping from external sources
- Key wrangling steps:
  - Removed incomplete or non-informative responses (e.g., missing values, "Prefer not to say")
  - Converted data types for numerical analysis (e.g., age ranges to numeric values)
  - Handled missing values using data imputation where appropriate
  - Normalized selected variables when needed
  - Split multiple-choice responses into separate rows for analysis (e.g., languages, databases)

1. Stack_Overflow_Survey_2024
2. https://www.kaggle.com/promptcloud/jobs-on-naukricom
3. Programming_languages_dataset

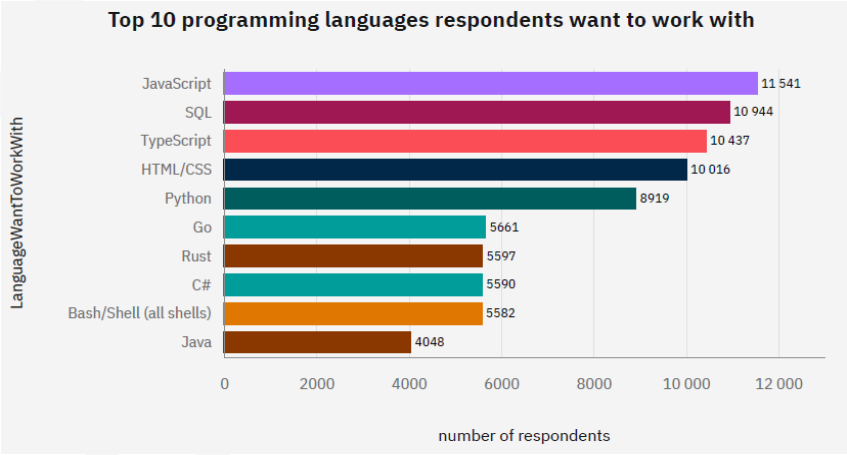
RESULTS

PROGRAMMING LANGUAGE TRENDS

Current & Past Usage



Future Interest



Quantitative Changes in Top 10 Programming Languages

In Top 10 Want to work with	Row Labels	Count of LanguageHaveWorkedWith	change between LanguageHaveWorkedWith and WantToWorkWith
YES	JavaScript	14943	
YES	SQL	12602	
YES	HTML/CSS	12410	
YES	TypeScript	10709	
YES	Python	9590	
YES	Bash/Shell (all shells)	7244	
YES	C#	6340	
YES	Java	5982	
NO	PHP	4644	-39,8%
NO	PowerShell	3438	-39,7%
		Top 10 total 87902	

Row Labels	Count of LanguageWantToWorkWith
JavaScript	11541
SQL	10944
TypeScript	10437
HTML/CSS	10016
Python	8919
Go	5661
Rust	5597
C#	5590
Bash/Shell (all shells)	5582
Java	4048
	Top 10 total 78335

LanguageHaveWorkedWith and WantToWorkWith		In Top 10 have worked with?
JavaScript	-22,8%	YES
SQL	-13,2%	YES
TypeScript	-2,5%	YES
HTML/CSS	-19,3%	YES
Python	-7,0%	YES
Go	71,3%	NO
Rust	145,1%	NO
C#	-11,8%	YES
Bash/Shell	-22,9%	YES
Java	-32,3%	YES

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Note: Multiple responses were allowed. Counts represent the number of respondents selecting each language, therefore totals may exceed the total sample size (N = 18,846).

PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

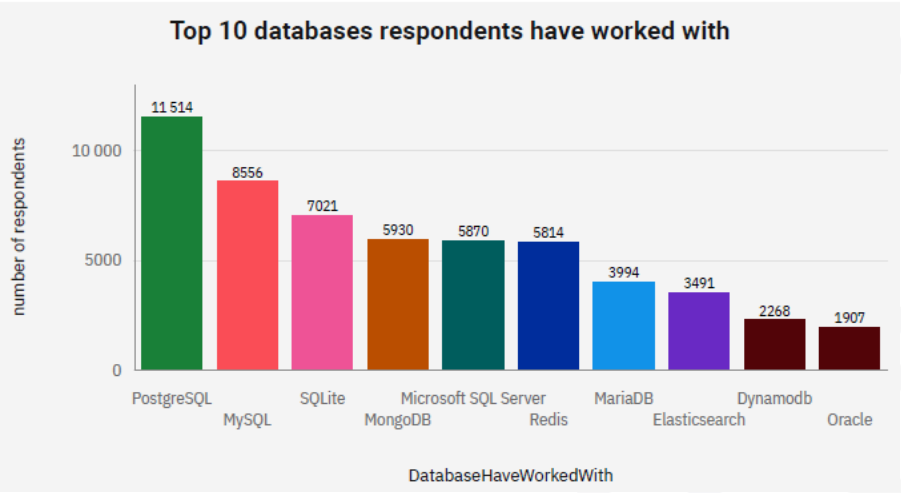
- JavaScript maintains its dominant position in both current usage and future interest rankings with 79.3% of respondents reporting they have worked with this programming language
- The number of respondents who want to work with **PHP** and **PowerShell** is nearly 40% lower than those who have worked with these languages
- The number of respondents interested in **Go** and **Rust** is significantly higher (by 71.3% and 145.1% respectively) compared to current usage

Implications

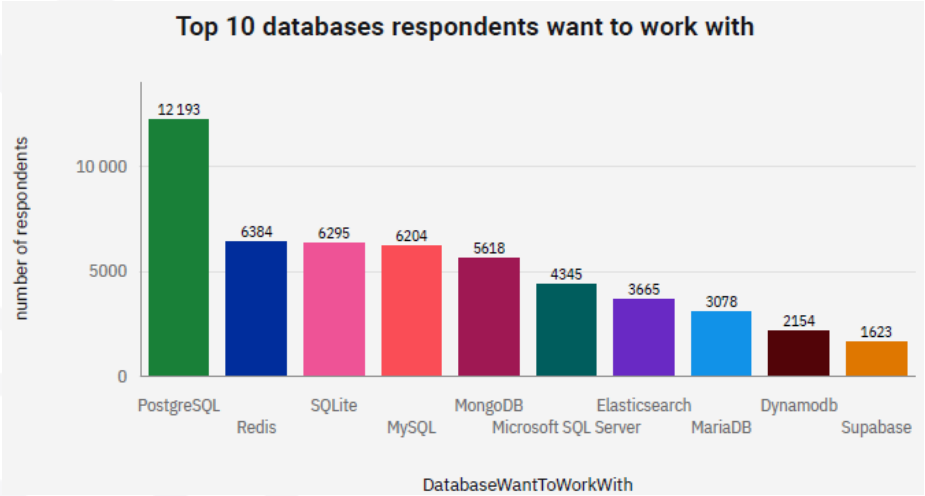
- There may be an increasing interest in Rust and Go among developers, which could lead to a growth in the number of specialists in these technologies
- Developers appear less willing to work with PHP, PowerShell, and Java, which may result in a potential shortage of specialists if market demand remains stable
- Technologies such as SQL, Python, TypeScript, and C# appear relatively stable in terms of both usage and interest

DATABASE TRENDS

Current & Past Usage



Future Interest



Quantitative Changes in Top 10 Databases

In Top 10 Want to work with?	Row Labels	Count of DatabaseHaveWorkedWith	change between DatabaseHaveWorkedWith and WantToWorkWith
Yes	PostgreSQL	11514	
Yes	MySQL	8556	
Yes	SQLite	7021	
Yes	MongoDB	5930	
Yes	Microsoft SQL Server	5870	
Yes	Redis	5814	
Yes	MariaDB	3994	
Yes	Elasticsearch	3491	
Yes	Dynamodb	2268	
No	Oracle	1907	-41,6%
		Top 10 total	
		56365	

In Top 10 have worked with?	Row Labels	Count of DatabaseWantToWorkWith	change between DatabaseHaveWorkedWith and WantToWorkWith
Yes	PostgreSQL	12193	5,9%
Yes	Redis	6384	9,8%
Yes	SQLite	6295	-10,3%
Yes	MySQL	6204	-27,5%
Yes	MongoDB	5618	-5,3%
Yes	Microsoft SQL Server	4345	-26,0%
Yes	Elasticsearch	3665	5,0%
Yes	MariaDB	3078	-22,9%
Yes	Dynamodb	2154	-5,0%
No	Supabase	1623	44,7%
		Top 10 total	
		51559	

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Note: Multiple responses were allowed. Counts represent the number of respondents selecting each database, therefore totals may exceed the total sample size (N = 18,846).

DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- PostgreSQL is the most popular database among developers in both current usage and future interest rankings
- Oracle, MySQL, Microsoft SQL Server, and MariaDB show a noticeable decline in developer interest compared to current usage, with **Oracle** experiencing the largest decrease (−41.6%)
- **Supabase** shows strong growth in interest, with 44.7% more respondents wanting to work with it compared to current usage

Implications

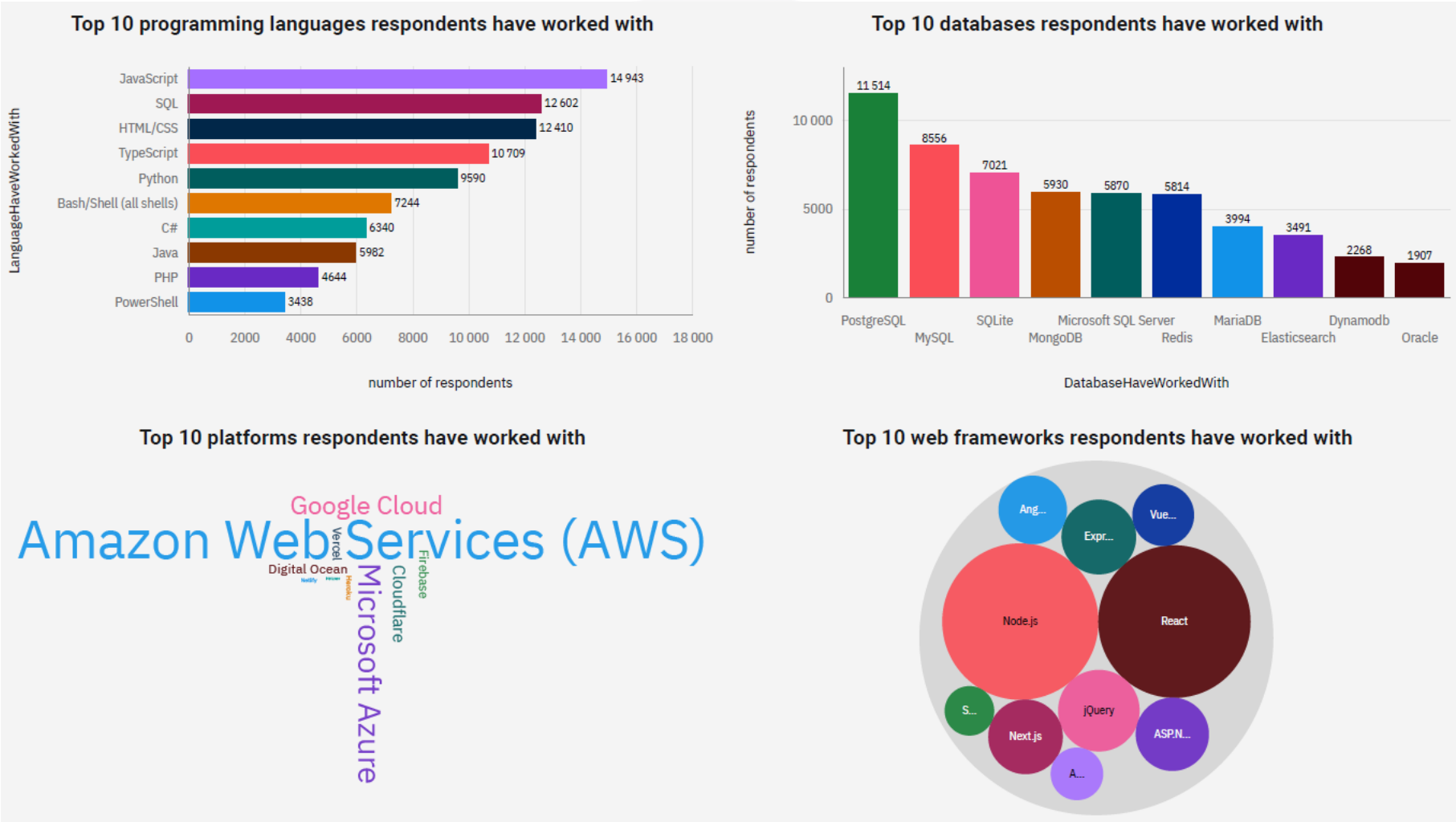
- PostgreSQL demonstrates a strong and stable market position, with both high current usage and even higher future interest, which may indicate continued growth
- Differences between current usage and future interest suggest that the most widely used databases have well-established positions in the market
- The significant decline in interest in Oracle may indicate a shift away from certain traditional or enterprise-focused database systems

DASHBOARD

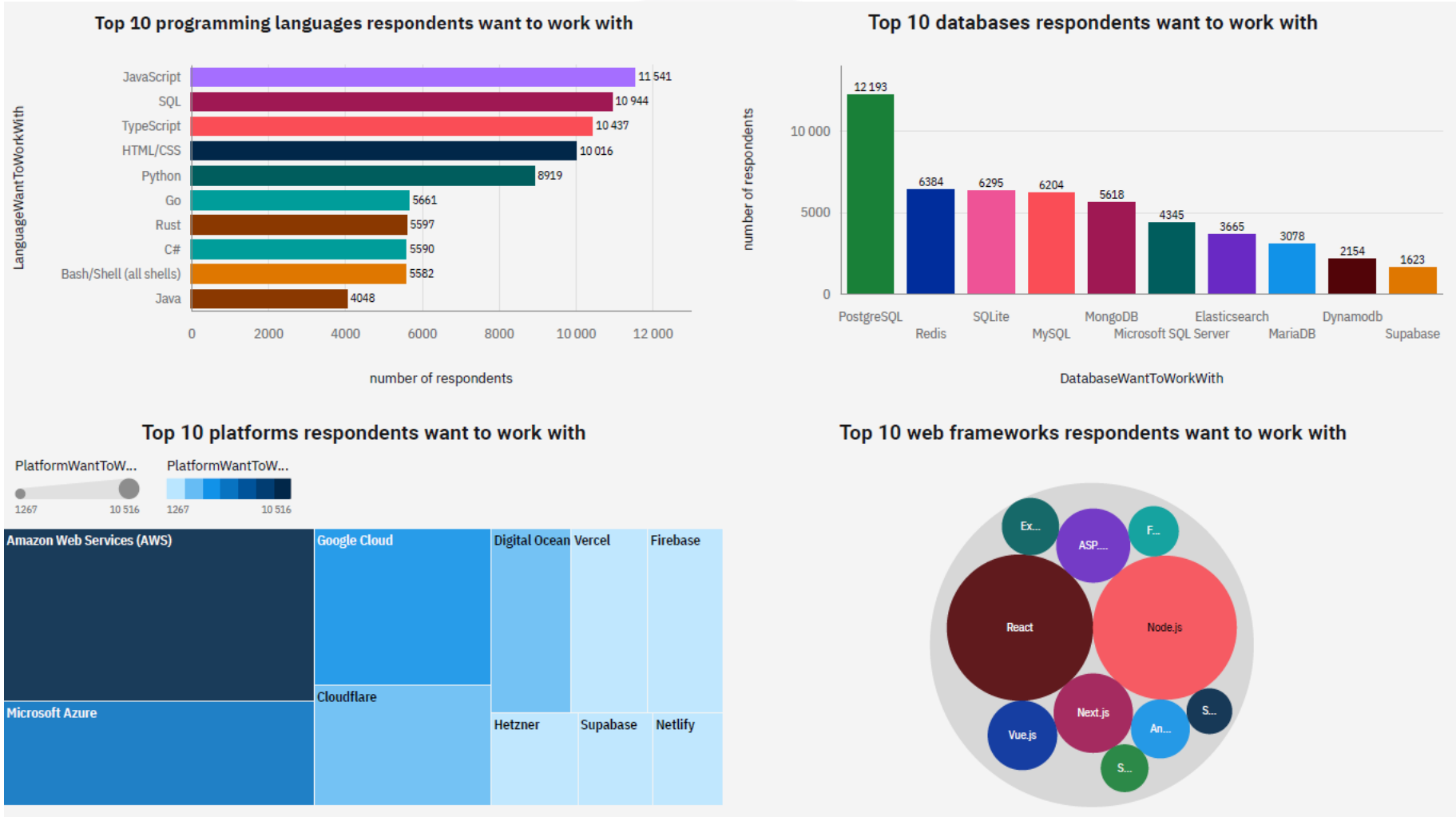
GitHub link of the Cognos dashboard =>

https://github.com/Damian4nT/IBM_capstone_project/blob/main/Technologies_%26_Demographics_dashboard.pdf

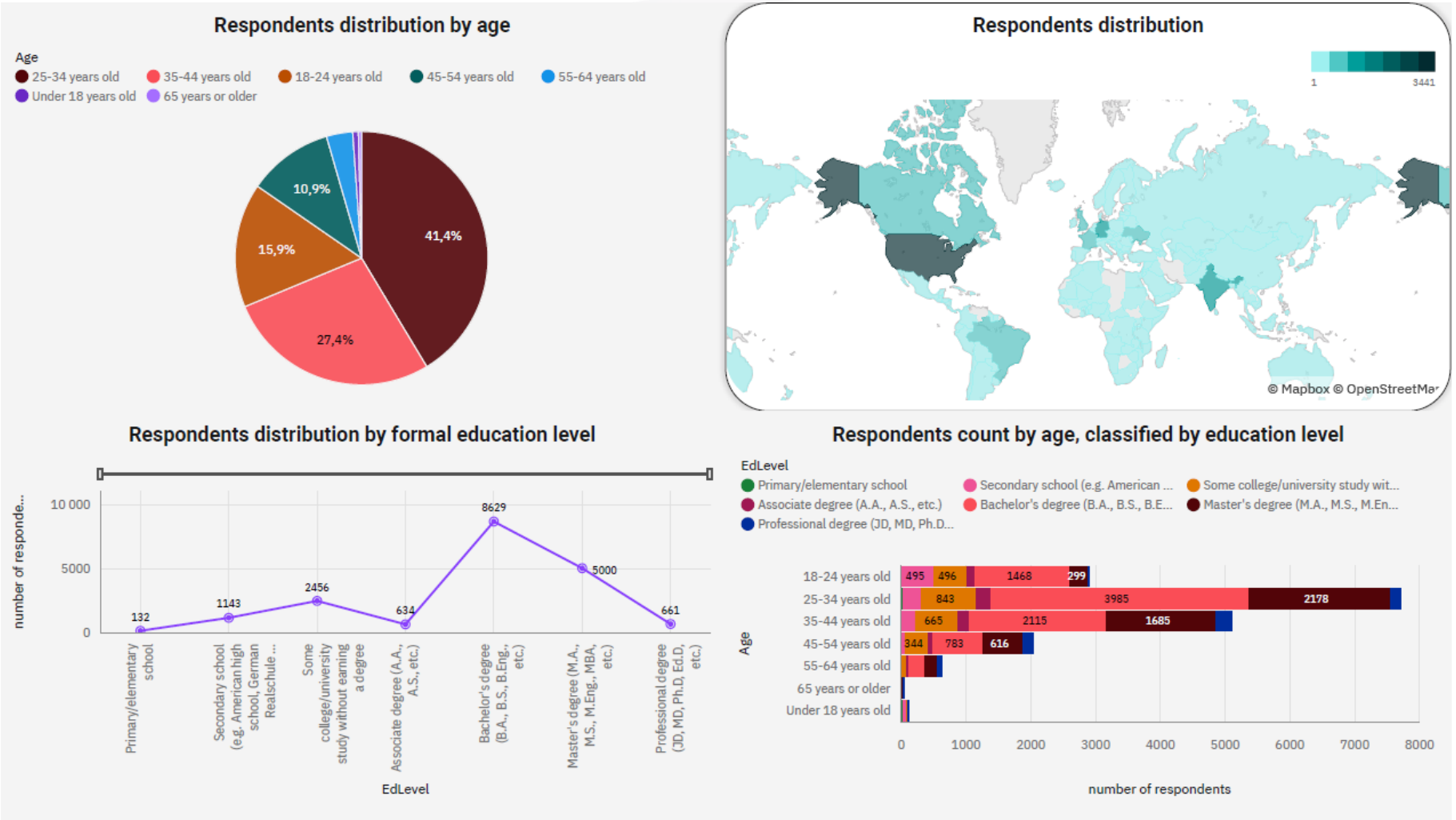
DASHBOARD TAB 1



DASHBOARD TAB 2



DASHBOARD TAB 3



DISCUSSION

OVERALL FINDINGS & IMPLICATIONS

Findings

- Most respondents are located in Europe and North America
- Nearly 85% of respondents are between 18 and 44 years old
- 93.2% of respondents have attended university or college
- Node.js(9,230 respondents) and React(8,999 respondents) are two dominant web frameworks in both current usage and future interest
- Amazon Web Services (AWS) maintains the leading position in both current usage and future interest

Implications

- Experience with university or college education appears to be a common entry path into the developer world
- Companies focusing on the most popular platforms may have access to a significantly larger pool of developers — for example, AWS is used by over 57.7% of respondents, while the 10th-ranked platform (Hetzner) is used by only 7.4%
- The concentration of respondents in Europe and North America suggests that the survey results may be less representative of other regions

CONCLUSION

The analysis shows that the current technology landscape is both stable and highly concentrated around a relatively small set of dominant tools. Technologies such as JavaScript, SQL, HTML/CSS, TypeScript, Python, PostgreSQL, AWS, Node.js, and React not only lead in current usage but also maintain strong positions in future developer preferences.

At the same time, the data suggests that changes in the ecosystem are gradual rather than disruptive — most developers continue to rely on well-established technologies, while only a few emerging tools show signs of growing interest.

The results also highlight that the developer population is relatively young and highly educated, which may influence both technology adoption and future industry trends.

APPENDIX

All cloud platforms repondents have worked with

Row Labels	Count of PlatformHaveWorkedWith
Amazon Web Services (AWS)	10871
Microsoft Azure	6681
Google Cloud	5537
Cloudflare	3874
Digital Ocean	3170
Firebase	3025
Vercel	2875
Heroku	2002
Netlify	1662
Hetzner	1392
VMware	1341
Supabase	1009
Managed Hosting	817
Linode, now Akamai	816
OVH	779
Fly.io	759
Render	633
OpenShift	574
Oracle Cloud Infrastructure (O	571
Vultr	437
Databricks	388
PythonAnywhere	346
OpenStack	308
IBM Cloud Or Watson	226
Scaleway	216
Alibaba Cloud	206
Colocation	140
Grand Total	50655

All web frameworks respondents have worked with

Row Labels	Count of WebframeHaveWorkedWith
Node.js	9230
React	8999
jQuery	4810
Express	4408
Next.js	4399
ASP.NET CORE	4319
Angular	4038
Vue.js	3642
ASP.NET	3094
Spring Boot	2916
Flask	2817
Django	2752
WordPress	2696
FastAPI	2487
Laravel	2016
AngularJS	1648
NestJS	1569
Svelte	1478
Ruby on Rails	1275
Blazor	1256
Nuxt.js	910
Htmx	815
Symfony	781
Astro	742
Fastify	622
Phoenix	567
Deno	476
Strapi	469
Drupal	468
Gatsby	455
CodeIgniter	416
Remix	407
Solid.js	273
Yii 2	245
Play Framework	175
Elm	133
Grand Total	77803

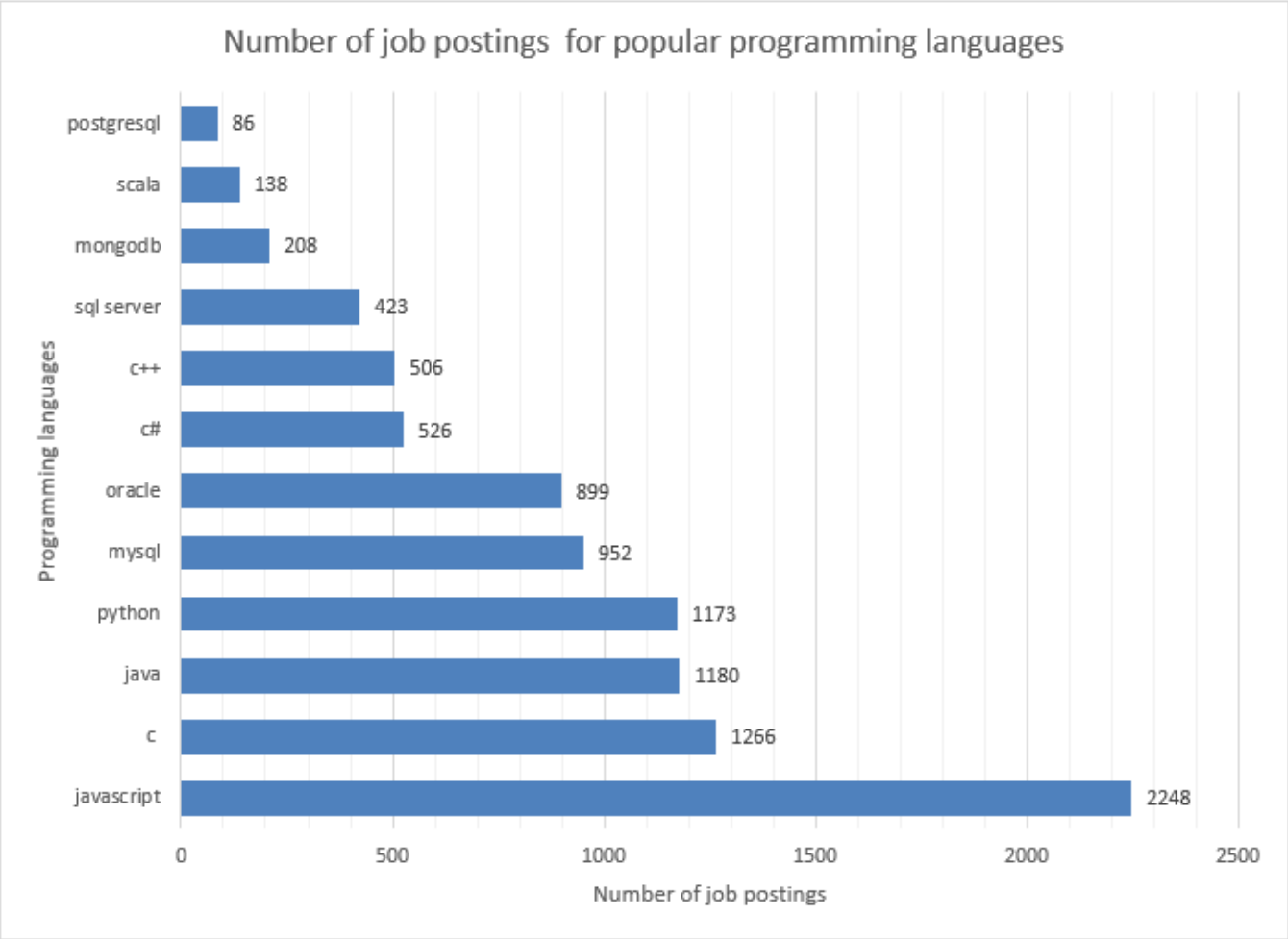
Top 30 countries by number of respondents

Row Labels	Count of Country
USA	3441
Germany	1339
India	1316
UK	1053
Ukraine	931
Canada	661
France	598
Brazil	587
Poland	494
Netherlands	480
Italy	452
Australia	444
Spain	409
Sweden	298
Austria	288
Switzerland	283
Czech Republic	217
Turkey	213
Belgium	184
Israel	174
Denmark	170
Mexico	169
Norway	167
Romania	166
Pakistan	151
Bangladesh	143
Indonesia	136
Portugal	136
New Zealand	132
Greece	127
Grand Total	**18845**

Europe and North America stats in top 30

Count: 22 Sum: 12276

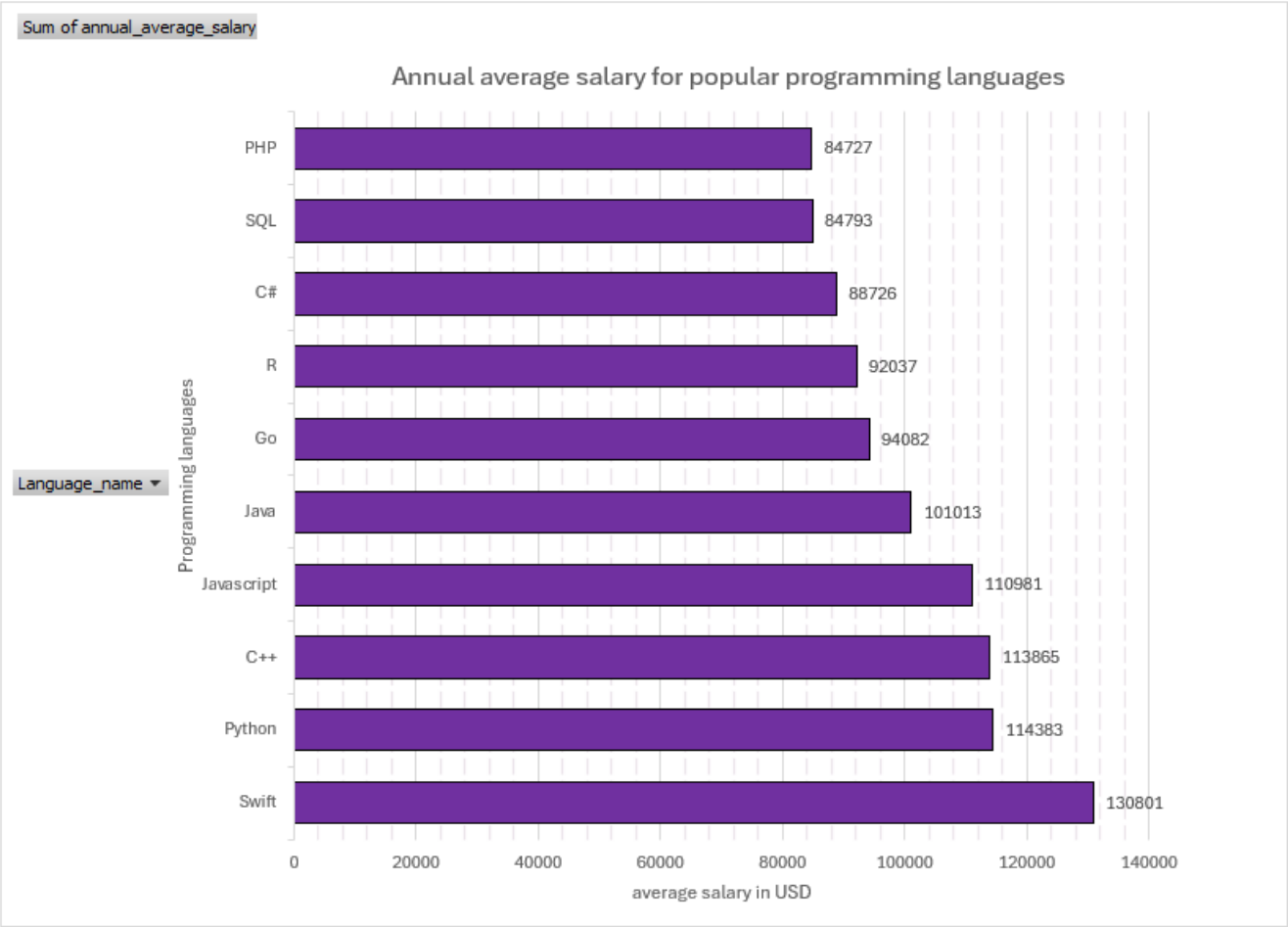
JOB POSTINGS



This chart is based on data collected via API. The data collection process is available here: https://github.com/Damian4nT/IBM_capstone_project/tree/main/Data_collection/Collecting_data_using_API

Note: The count for the "C" programming language may be slightly inaccurate due to limitations in text matching logic.

POPULAR LANGUAGES



This chart is based on data collected using web scraping. The data collection process is available here: https://github.com/Damian4nT/IBM_capstone_project/tree/main/Data_collection/Web%20Scraping